**NASA Human Exploration Rover Challenge - Project Proposal**

**UNIVERSIDAD CATÓLICA BOLIVIANA "SAN PABLO" Sede La Paz**

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College/University

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Remote-Controlled Rover Division(RC)

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## 0. Introduction

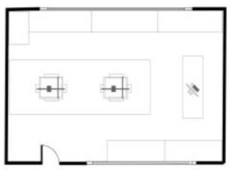
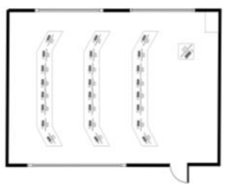
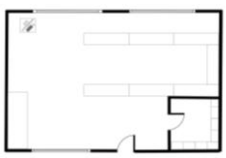
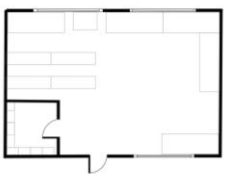
This proposal presents the development of a Remote Controlled (RC) Rover built by students of the Mechatronics Engineering (IMT) degree from the *Universidad Católica Boliviana "San Pablo" (UCB) Sede La Paz* for NASA Human Exploration Rover Challenge (HERC). Taking advantage of the technical resources offered by the university, the team aims to design, build, and develop a reliable RC vehicle for extreme terrain with subsystems to accomplish specific missions and tasks. The project not only seeks to foster teamwork and engineering implementation, but also prioritizes safety, sustainability, and educational outreach through STEM activities to inspire new generations in Bolivia.

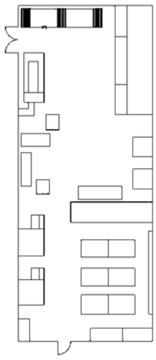
## 1. Facilities and Equipment

### 1.1 Facilities, Equipment, and Supplies for RC Vehicle Development

UCB's La Paz campus includes an Engineering Building (Block C) equipped with various laboratories for electronics, 3D designs, programming sessions, meeting rooms, and manufacturing areas. These areas have the necessary facilities, resources, and services, such as: water, electricity, Wi-Fi connectivity, restrooms, ventilation, smoke detectors, fire alarms, fire extinguishers, and accessible and safe infrastructure that has the necessary signage. Table 1 shows more precise descriptions of the spaces mentioned.

***Table 1. Description of the Work Areas.***

| N° | Work Areas | Room Diagram | Description |
|---|---|---|---|
| **1** | **Prototyping Lab (C101)** |  | **Area:** 322.9 $ft^2$<br>**Capacity:** 15 people<br>**EQ/tools:** 21 3D FDM printing machines; 4 3D SLA printing machines; 2 PCB manufacturing device<br>**Availability:** 20 hours/week<br>**Location:** 2nd floor of Block C |
| **2** | **Computing Laboratories** |  | **Area:** 699.6 $ft^2$<br>**Number of laboratories:** 5<br>**Capacity per laboratory:** 35 people<br>**EQ/tools:** 30 PCs with licensed software (SolidWorks 2024, Altium Designer 2024, MATLAB R2023B, Fusion 360)<br>**Availability:** 50 hours/week (this total represents the combined available hours of all laboratories).<br>**Location:** 1st floor of Block C |
| **3** | **Electronic Systems Lab (C202)** |  | **Area:** 968.8 $ft^2$<br>**Capacity:** 30 people<br>**EQ/tools:** 19 DC power source, 19 Function generators; 18 Three-phase machine; 18 Digital multimeter; 18 Digital oscilloscope<br>**Availability:** 49 hours/week<br>**Location:** 2nd floor of Block C |
| **4** | **Scientific Society (SCE-IMT) Lab (C203)** |  | **Area:** 1,022.6 $ft^2$<br>**Capacity:** 30 people<br>**EQ/tools:** Electronic stuff; 14 Work tables<br>**Availability:** 70 hours/week<br>**Location:** 2nd floor of Block C |

| | | | |
|---|---|---|---|
| **5** | **Machine and manufacturing floor (CS203)** |  | **Area:** 1,506.9 $ft^2$<br>**Capacity:** 30 people<br>**EQ/tools:** Equipment for welding, milling, turning, drilling, grinding, sanding, polishing, laser cutting, angle grinder - BOSCH, plasma cutter station - TAURO, MIG / MAG welding station - TAURO, TIG welding station - TAURO, CNC milling machine - EMCO, oxyacetylene welding kit - VICTOR, BOSCH crosscut saw, drill bench - KAILI, hammer j8 drill - DEWALT, carbon dioxide tank - PRAXAIR, gasogen tank - DUPLEX, oxygen tank - COBOXI, CNC lathe - EMCO and a Laser cutting machine of 100W<br>**Availability:** 48 hours/week<br>**Location:** 2nd basement of Block C |

Considering the work required to complete the RC vehicle, the labs will be used as follows:

1. **Prototyping Lab (C101):** Designed lab to fabricate all components due to the location and availability of additive manufacturing systems. Technical training in operating 3D printers is required, to be conducted by a specialized additive manufacturing technician as necessary personnel. This area is available between 9:00 am and 1:00 pm only on working days.
2. **Computing Lab:** Designated for computer aided design (CAD) modeling, SolidWorks simulation also and embedded systems programming sessions. Training in parametric 3D design with SolidWorks CAD and embedded software development is required, to be conducted by a mechanical design engineer and embedded systems engineer as necessary personnel. There are a lot of hours available because the university provides four of these labs, and each one manages almost the same count of students.
3. **Electronic Systems Lab (C202):** This lab contains the majority of electronic instrumentation and equipment to develop the electronic subsystems. Technical training in electronic circuits analysis and instrumentation for the development of electronic systems is required, to be conducted by an embedded systems engineer as necessary personnel.
4. **Scientific Society Lab (C203):** Lab C203 will serve as a collaborative workspace for prototype assembly. This space is designated for SCE-IMT projects so it will be available from 7:15 am to 9:00 pm. Training in project coordination, meeting facilitation and organizational management, to be conducted by a lab coordinator as necessary personnel.
5. **Machine and manufacturing floor (CS203):** If needed, this lab contains industrial grade machining equipment for the fabrication of rover parts. Training in the operation of machine tools and manufacturing processes is required, to be conducted by a mechanical engineering technologist as necessary personnel.

These laboratories are supervised by full-time faculty from the UCB Mechatronics Engineering program. Their role includes delivering technical advice, overseeing equipment operation compliance and enforcing adherence to laboratory safety standards.

### 1.2 Institutional Support and Sponsorship

The RC rover team represents the UCB's La Paz branch in the HERC competition. It is expected that the university provides its facilities, including: infrastructure, laboratories, and equipment, such as CNC machines, some of the 3D printers needed, and others, for the prototype construction as mentioned in point 1.1. In addition to the support of the institution, the project also has the support of the Student Scientific Society of Mechatronics Engineering (SCE-IMT).

The benefits offered by this society are:

- A fund/budget specially allocated to the project, which will help acquire/buy components and various elements for robot's assembly.

- Contacts with external institutions and research projects.
- Support and development of STEM activities.
- Progress review meetings monitoring hardware, software, teamwork, documentation, operation, planning, and STEM Engagement.

## 2. Safety

Priority will be given to equipment safety through clear safety standards established from the outset of the project. These standards include protocols for machinery, the use of electricity, personal protective equipment (PPE) such as eye protection, insulated gloves, coveralls, and the organization of the workspace based on ISO standards. A safety officer, who has formal training in occupational health and safety, will oversee their communication and implementation. Compliance will be mandatory during all phases of the project, which will also be formally addressed in the project plan.

***Table 2: Hazards and how to prevent them in all phases of work.***

| N° | Danger/Risk | Phase | Risk Level | Security Measure |
|---|---|---|---|---|
| **1** | **Short circuit / Electric shock** | Construction | EXTREME | Insulated wiring; grounded shielding; fail-safe systems |
| **2** | **Risk of physical injury** | Construction | HIGH | Mandatory use of PPE (eye protection, insulated gloves, coverall, dust coat); organized and safety ergonomics workspace |
| **3** | **Hazard fumes produced during welding and 3D printing operations.** | Construction | HIGH | Ventilated workspace |
| **4** | **Hard and sharp edges** | Finishing | HIGH | Use non-hazardous materials, avoid designing them, and chamfered of edges |
| **5** | **Involved risk of lithium batteries (fire, explosions, toxic fumes, etc)** | Finishing | EXTREME | Impact-protected areas and a fan in the battery compartment, and implementation of temperature sensors |
| **6** | **External damage to exposed areas** | Finishing | MEDIUM | Enclosure for electrical and vulnerable components |
| **7** | **Vehicle hits and collisions** | Operation | HIGH | Remote and local shutdown switches; delimitated safety zones; visual warnings |
| **8** | **Disconnection between controller and RC Rover** | Operation | HIGH | Periodic connection check via software, and emergency stop if an error is detected |

***Table 3: Project Safety Measures by Phase and Location***

| Project Phase | Location/Facilities | Safety Measures |
|---|---|---|
| **Design** | **Computing Lab (C102):** PCs with design software: SolidWorks, Fusion 360, MATLAB, Proteus. | - CAD and simulation to identify hazards and analyze stress strength.<br>- Eliminate pinch points and sharp edges in models.<br>- Selection of insulated, grounded and fail-safe components. |
| **Fabrication** | **Prototyping Lab (C101):** 3D FDM and SLA printers, PCB devices. | - Mandatory PPE (eye protection, insulated gloves, coverall, dust coat) |

| | | |
|---|---|---|
| | **Scientific Society Lab (C203):** Workbench, ventilation, fire safety.<br>**Manufacturing Floor (CS203):** Welding, milling, turning, grinding, CNC machines; ventilation, fire safety equipment. | and tool handling<br>- Ventilation for welding and 3D printing<br>- Restricted access to high-risk machinery<br>- Supervised integration and wiring of components |
| **Storage** | All labs and storage rooms in the University's Engineering Block include ventilation, fire alarms, and extinguishers. | - Fire-resistant, ventilated enclosures for lithium-ion batteries.<br>- Institutional fire safety measures<br>- Clearly labeled and separated areas |
| **Testing** | Prototyping Lab, Mechatronic Lab and Machine Shop: Use of specific test areas with safety measures. | - Delimited test areas with signage<br>- Remote or local shutdown systems<br>- Emergency stops<br>- Operator communication<br>- Structural and functional checks before testing. |

In accordance with safety regulations, the team has identified the electrical and electronic materials that may involve chemical substances or potential hazards during handling (e.g., solder, flux, cleaning solvents, thermal compounds, and lithium-ion batteries). For these items, preliminary Safety Data Sheets (SDS) are provided as references.

The team will consider **safety implementation during all phases as a crucial part of the project** by applying standardized practices while implementing the design, fabrication, testing, and operation phases:

- **Design/Fabrication Phase:** When it comes to the embedded design, specific parts will be protected with impact resistant compartments with appropriate ventilation in order to protect the subsystems in case of physical damage or overheating.
- **Testing Phase:** For this specific stage of the project, team members will follow strict safety practices, including the use of proper Personal Protective Equipment (PPE) depending on the workspace, and clear labeling of tools and materials.
- **Operation/Competition Phase:** Testing before the competition will take place in specific safety zones, this procedure includes regular inspection of the Rover to ensure parts remain safe and fixed. During competition, the team members in charge of RC control will have the necessary safety requirements, from the prior training described in Table 4 to the necessary equipment mentioned in Table 5.

Furthermore, the following table describes the proper development and exchange of information related to the safe use of the various software, tools and machines used in the mechanical and electrical construction of the RC Rover.

***Table 4: Description of Student Briefings***

| N° | Subject Description | Timeframe | Personnel | Description |
|---|---|---|---|---|
| **1** | Project Security Fundamentals and Organizational Rules | Week 1 (October) | Team leader, chief mechanic and all design and simulation team members | This is a mandatory introductory session. It will cover the team's safety philosophy, general laboratory rules (C101, C203, CS203), and an overview of the general safety plan. |
| **2** | Safe Design in CAD: Virtual Hazard Identification | Week 1 (October) | Chassis manager, wheel manager, steering manager, mechanical design assistant, | Directly linked to the Design phase in lab C102. It will teach students how to use SolidWorks/Fusion 360 not only to design, |

| | | | | |
|---|---|---|---|---|
| | | | drivetrain manager, Task 2 and Task 3 manager | but also to identify and eliminate pinch points, sharp edges, and snag points in their models before manufacturing them. |
| **3** | Selection and Proper Use of Personal Protective Equipment (PPE) | Week 2 (October) | Chassis manager, drivetrain manager, chief mechanic, mechanical construction assistant, management manager | Practical session that complements all the others. It will show which PPE (glasses, gloves, long sleeves, closed shoes) is mandatory for each task and how to use it correctly. |
| **5** | Ventilation and Chemical Risk Management | Week 2 (October) | Chassis manager, wheel manager, drivetrain manager, chief mechanic, mechanical construction assistant, management manager | Focused on the risk of hazard fumes. It will teach you how to set up and use localized extraction systems in labs for welding and 3D printers, explaining the specific dangers of fumes and resins. |
| **6** | Soldering | Week 3 (October) | Chassis manager, drivetrain manager, chief mechanic, mechanical construction assistant, management manager | Addresses electrical hazards in the Manufacturing and Testing phases. Teaches welding techniques with ventilation, use of multimeters, insulated and grounded wiring. |
| **7** | Safe Handling, Storage, and Charging of Lithium Batteries | Week 3 (October) | Head of Embedded | It will cover specific protocols for storage in fire-resistant enclosures, supervised loading, and emergency response (leak, inflation, fire). |
| **8** | Emergency and Incident Response Protocol | Week 3 (October) | Team leader, chief mechanic and all design and simulation team members | The Safety Officer will teach the entire team what to do in the event of an accident: from a minor cut to an electrical or battery fire. This will include the use of first aid kits and the procedure for reporting incidents for continuous improvement. |

***Table 5: Required PPE for Each Phase of the Project***

| Project Phase | PPE | Objective |
|---|---|---|
| **Design and Fabrication** | Safety goggles, eye protection, hearing protection, insulated gloves, safety footwear, and coveralls when using tools/equipment/machines such as grinders, drills, or welding stations. | Prevent cuts, burns and debris injuries. |
| **Testing** | Insulated gloves, flame resistant coveralls, and safety-toe footwear. | Protect the team from mechanical and electrical hazards. |
| **Operation and competition** | Team members working directly with the robot will wear fire-resistant coveralls, task-specific gloves, protective helmets with integrated eye protection, knee pads, elbow pads, and safety-toe footwear, while the rest of the team will maintain a safety perimeter around the competition area. | Ensure safe driving/controlling and protect team members and outsiders. |

**Note:** PPE availability will be checked by the safety officer at the start of each work session.

## 3. Technical Design

### 3.1 Rover Design Overview

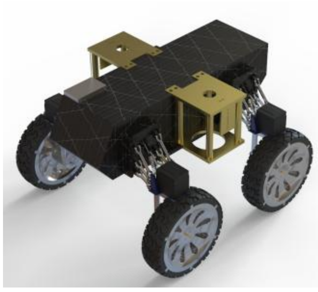


**Fig 1. Conceptual Rover Design**

The overall design concept focuses on developing a remote-controlled off-road vehicle capable of moving through various obstacles without getting stuck. Embedded architecture complies with HERC guidelines by separating mobility and task controls: a commercial RC unit provides primary drive control, while a custom-built controller manages task operations. A Raspberry Pi 5 serves as the central processor for telemetry and data logging, connected via CAN bus to three Raspberry Pi Pico W units. Two Picos control the drive subsystems with PID-based velocity measurement, and the third manages the crew area (GPS, temperature, G-forces) along with task tools. Two 14.8V batteries ensure 30 minutes of continuous operation, approximately, with 258.6 watts of maximum power consumption. Mechanical design (Figure 1) focuses on four key features: off-road capability, rugged suspension, lightweight construction, and excellent steering. These requirements led to the choice of a four-wheel vehicle design, with the entire design process geared toward ensuring that this RC rover would perform reliably in challenging terrain. The chassis is designed to be built entirely using additive manufacturing processes, incorporating a crew area in the front part, independent suspension to ensure contact [1] and transmission with steering systems for each wheel.

### 3.2 Wheel Design


**Fig 2. Wheel Conceptual Design**

The wheel design is inspired by a commercial all-terrain design that enables remarkable performance on difficult terrain. It consists of a modular design: the inner part of the wheel will be made of acrylonitrile styrene acrylate (ASA), and the outer part of thermoplastic polyurethane (TPU), both produced via 3D printing. The rim, made of ASA, maintains a conventional rigid shape to be assembled onto a motor, while the tread, made of TPU, has a hollow structure with internal supports that allow the wheel to flex upon ground contact. Traction relies on the deep grooves of the wheel, based on commercial all-terrain tires, ensuring proper handling on various surfaces such as dirt, mud, or snow.

### 3.3 Drivetrain Concept and Design with Fabrication Plans

The powertrain will include transmission, suspension, and steering subsystems. The manufacturing plan will begin with a complete CAD design using SolidWorks software, including stress and motion simulations. A prototype will then be built using polylactic acid (PLA) through 3D printing to perform the necessary tests. The third step of the construction plan will be to evaluate the performance of the prototype to improve it in a later version. Therefore, once the final prototype has been validated, it will be printed using technical materials such as ASA and TPU, thanks to their various structural advantages [2], along with the implementation of electronic components. Therefore, the aim is for each wheel to have a drive train mounted directly on it, which involves an individual motor, suspension with independent shock absorbers, and a steering system based on a servomotor. In this way, through prototyping and subsystem integration, the construction of a radio-controlled vehicle that meets the proposed max speed of 5 mph and a torque capable of overcoming the proposed obstacles will be completed.

### 3.4 Tasks

**Task 1 (Soil Sampling):** To address the challenge of soil sampling, which involves extracting at least 15 ml from a 3-inch diameter, 4 inch deep, circular container buried at ground level, the team propose using a helical drilling system mounted on the RC rover chassis. The system's features allow the rover to penetrate the soil, break up the moist soil, and transport the material in a controlled manner to a removable container mounted on the vehicle. The container can be easily removed for emptying and subsequent reporting and prevents spills during subsequent tasks and obstacles. The main advantage of this system is that it ensures continuous and controlled soil capture, optimizing storage and reducing the risk of clogging thanks to its auger-based helical design.

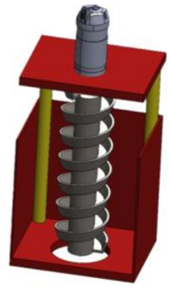


**Fig 3. Proposed Mechanism for Task 1**

**Task 2 (Water Sampling):** To address the challenge of task two, a retractable extraction system located on the side of the vehicle chassis is proposed. The extraction system includes a threaded lowering mechanism, driven by a DC motor, which moves the hose along with the pump until it reaches the surface of interest. Once in contact, the electric pump extracts the sample, which is stored in a container integrated as a subsystem within the same structure of the task. This container is designed with mechanical features that prevent spills and protect the vehicle's internal circuits. Inside is the pH sensor, which allows precise sample measurements. Finally, the data recorded by the sensor is transmitted to the central CPU, optimizing the data acquisition and analysis process.

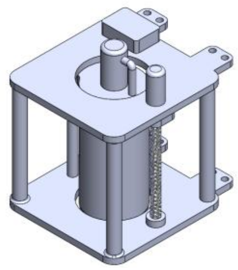


**Fig 4. Proposed Mechanism for Task 2**

**Task 3 (Air Sampling):** For this task, it was decided to use the MG-811 $CO_2$ sensor, since it is easier to obtain in Bolivia. The sensor will be placed inside a small chamber at the front of the rover, protected by the bumper. The chamber will have an inlet and an outlet aligned with the airflow, connected to the container through a short silicone tube with a dust filter and, if necessary, a check valve. This setup protects the sensor from shocks and turbulence, shortens the response time, and allows air sampling from mid-height inside the container, reducing errors from stratification and ensuring consistent measurements during operation.

***Table 6. Description of Expected Technical Challenges and Possible Solutions.***

| Technical Challenges | Consequences | Solutions |
|---|---|---|
| **Vibrations and friction with high-speed motors** | Component war, noise, reduced service life | Sealed bearing; structural reinforcements; high viscosity grease; dynamic engine and wheel balancing |
| **Energy loss or insufficient torque** | Poor performance, overheating, reduced autonomy | Choose motors with sufficient torque and use 4-cell 12v LiPo batteries. |
| **Vulnerability and mounting issues** | Structural instability, risk of damage, difficulty of maintenance | Protective casing is integrated into the suspension. Alternative: Internal motors with torque transmission via shafts, shock absorbers and damping materials to reduce stress concentration |
| **Protection against dust, sand, and moisture** | Wear, short circuits, mechanical jams, corrosion. | Isolation of control systems, circuits and use of protective covers. |
| **Terrain limitations (rocks, slopes, mud)** | Traffic jams, overturns, damage to the chassis and work tools | Articulated suspension, all-terrain wheels and adequate chassis height. |

## 4. Project Plan

The following Gantt chart describes the project schedule from project initiation to the final product considering three workstreams: embedded systems development, mechanical design and fabrication, and STEM division.

***Table 7. Gantt Chart of the Project.***

| GANTT CHART | | OCT | | | | NOV | | | | DEC | | | | JAN | | | | FEB | | | | MARCH | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **AREA** | **TASK** | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| STEM | Stem activities | | | | | | | | | | | | | | | | | | | | | | | | |
| Entire team | Project definition | | | | | | | | | | | | | | | | | | | | | | | | |
| Embedded team | General design | | | | | | | | | | | | | | | | | | | | | | | | |
| Mechanical team | General design | | | | | | | | | | | | | | | | | | | | | | | | |
| Embedded team | DR Redaction | | | | | | | | | | | | | | | | | | | | | | | | |
| Mechanical team | DR Redaction | | | | | | | | | | | | | | | | | | | | | | | | |
| Embedded team | Acquisition of material | | | | | | | | | | | | | | | | | | | | | | | | |
| Mechanical team | Acquisition of material | | | | | | | | | | | | | | | | | | | | | | | | |
| Entire team | Welding and assembly | | | | | | | | | | | | | | | | | | | | | | | | |
| Mechanical team | Tool construction | | | | | | | | | | | | | | | | | | | | | | | | |
| Entire team | Testing endurance and mobility | | | | | | | | | | | | | | | | | | | | | | | | |
| Embedded team | Testing sensors | | | | | | | | | | | | | | | | | | | | | | | | |
| Mechanical team | Paint and exterior details | | | | | | | | | | | | | | | | | | | | | | | | |
| Mechanical team | Test on RC | | | | | | | | | | | | | | | | | | | | | | | | |
| Entire team | ORR redaction | | | | | | | | | | | | | | | | | | | | | | | | |
| Entire team | Final test | | | | | | | | | | | | | | | | | | | | | | | | |
| Entire team | Field and obstacle training | | | | | | | | | | | | | | | | | | | | | | | | |

***Table 8. Risk Matrix and Mitigation Plan***

| N° | Description | Category | Probability | Impact | Mitigation |
|---|---|---|---|---|---|
| 1 | Burnout and damage of electronic components | Technical | Medium | High | Get spare parts |
| 2 | Structural damage of the robot chassis | Mechanical | Medium | High | Structural damage will be identified for replacement. |
| 3 | Insufficient battery power | Technical | Medium | High | If we cannot find a battery that meets our needs, we plan to use a low-power circuit. |
| 4 | Software comm failure | Embedded | Low | High | Redundant comm protocols and a lot of pre-testing. |
| 5 | Team member injury | Safety | Low | High | First aid |
| 6 | Limited budget for spare parts | Budget | Medium | Medium | Planning procurement early and seek sponsors |

*Table 9. Materials and Tools the Team Has on Hand.*

| Category | Construction tools | Power and electrical components | Electronics and control | Mechanics and actuators |
|---|---|---|---|---|
| **Materials & Components the team has on hand** | Basic Tools: Toolbox, screwdrivers, Allen keys, wrenches | LiPo Batteries | Programmable professional remote control | servo TD8120 |
| | Drilling & Cutting: Drill with bits, grinder | Wires | ESP32 | servo TD8120MG |
| | Soldering iron, multimeters, PCB manufacturing device | 12V Fan | Raspberry Pi | 4 Aluminum shock absorbers (4.72 in) |
| | Welding: Electric/arc welder | 12V Buzzer | Basic Electronics Components | REV HD HEX Motors |
| | 3D Printing: 3D printer, TPU filament | Step-down Module | WT901C-232: 9-axis IMU | Assorted Screws |
| | Safety Equipment: PPE (gloves, overalls, industrial boots) | Motor Drivers | Turnigy 9X 9Ch Mode 2: 9-channel transmitter | Angle Brackets |

*Table 10. Materials, Supplies, and Travel to be Purchased*

| Category | Item | Distributor | Quantity | Unit price (USD) | Total (USD) |
|---|---|---|---|---|---|
| **Materials & Components the team need to purchase** | Raspberry pi pico W | Epy Electronics, Bolivia | 6 | 27.50 | 165 |
| | Electronic components | EC Robotics, Bolivia | Many | 0.80 | 20 |
| | Batteries, regulators, etc. | EC Robotics, Bolivia | 3 sets | 70 | 210 |
| | ASA Carbon Fiber Filament | Amazon | 4 | 28.79 | 115.16 |
| | Siraya Tech TPU 85A, ISO 10993 | Amazon | 4 | 33 | 132 |
| | Suspension shock absorbers | Locomotion RC hobbies, Bolivia | 4 | 16.16 | 64.64 |
| **Travel & Logistics (5 days)** | Plane ticket LPZ – MIA – LPZ | BOA | 17 | 980 | 16660 |
| | Transport in city | KAYAK, Miami | 1 | 945 | 945 |
| | Food and incidentals | Fast Food Restaurants | 17 | 200 | 3400 |
| | Rover transportation | BOA | 1 | 100 | 100 |
| | Accommodation | Booking | 17 | 600 | 10200 |
| | Travel insurance | Generic Prices | 17 | 80 | 1360 |
| **STEM Engagement Activities** | Stationery for 10 workshops | Local suppliers, Bolivia | 10 | 20 | 200 |
| | Snacks for 10 workshops | Local suppliers, Bolivia | 10 | 22 | 220 |
| | Intercity bus tickets for 5 members | El Dorado | 5 | 29 | 145 |
| | Local transportation for 4 members | Local Transport companies | 4 | 25 | 100 |
| **TOTAL** | | | | 2002,16 | 34036,80 |

To successfully develop and present the teleoperated rover project, we aim to gather the necessary financial and logistical support from internal and external sources. Figure 5 includes realistic actions focused on resource management, sponsorship outreach, and basic contingency preparation.

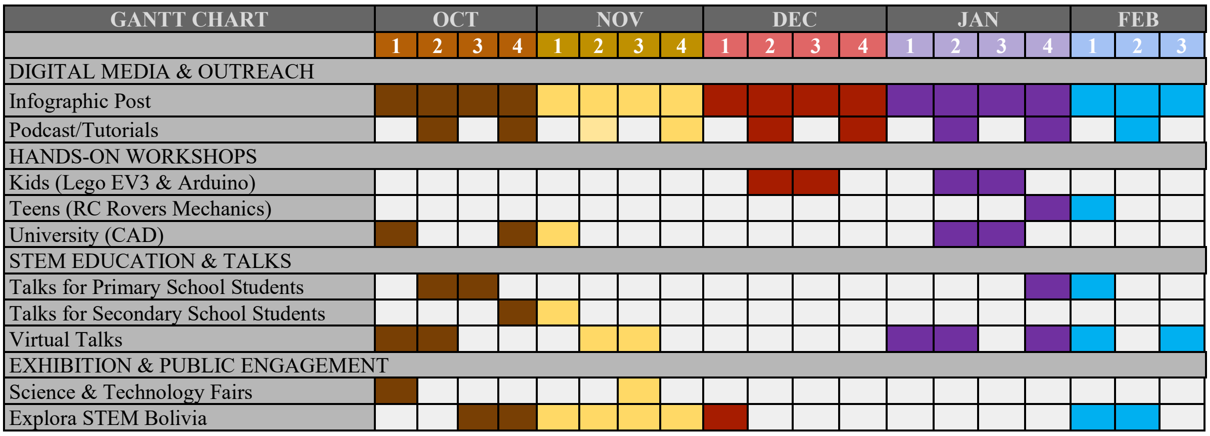

**Figure 5. Project Funding Summary**

## 5. (Community) STEM Engagement Plan

The team designed a STEM Community Engagement Plan to support their participation in NASA HERC and inspire students in Bolivia. The plan mixes workshops on robotics, embedded systems, and CAD with school talks, science fairs, and a traveling program called **"Explora STEM Bolivia: Aprende y Descubre"**, which will reach cities in Bolivia.

It is estimated that each workshop will bring together around 40 participants and that the activities in schools will involve more than 100 students per event, while the online content could reach between 1,000 and 2,000 views per month, with projected growth. Partnerships with Voces del Poder, Tecnonautas, and the Student Scientific Society of Mechatronics Engineering will expand coverage and strengthen these STEM efforts to generate a sustainable impact and motivate new generations toward science and technology.

***Table 11. Gantt chart of planned STEM activities***




## 6. Institutional Support Letter

The following page includes the official support letter from Universidad Católica Boliviana "San Pablo" – Sede La Paz, confirming institutional awareness and approval of the team's participation in the NASA Human Exploration Rover Challenge (HERC).

UNIVERSIDAD
CATÓLICA
BOLIVIANA

La Paz, August 21st, 2025

**Selection Committee,**
**NASA Human Exploration Rover Challenge 2026**
**U.S. Space & Rocket Center**
**Huntsville, Alabama, USA**

***Ref. Institutional Support Letter – NASA HERC 2026***

This letter certifies that the **Universidad Católica Boliviana "San Pablo"- Sede La Paz,** legal entity registered under **NIT 1020141023**, located in the city of La Paz, Bolivia, at Av. 14 de Septiembre N° 4807, is aware of and fully supports the participation of the **team SEEKER-HERC in the Remote Controlled Category of the University Division of the NASA Human Exploration Rover Challenge 2026.** Although the Universidad Católica Boliviana "San Pablo" shares the same name across its campuses in different cities, each campus operates independently. Therefore, this endorsement is exclusively made by the La Paz campus. The team belongs to the Mechatronic Engineering program and also includes members from other academic programs to promote STEM initiatives and outreach efforts towards this edition of the challenge.

The team is under the academic guidance of **MSc. Xavier Alexis Murillo Sánchez (C.I. N° 8437124 LP)** as Team Advisor of the project, and **Adrián Eduardo Vargas Llanquipacha (C.I. N° 13721258 CH)** as Team Leader. It consists of a total of **17 members**.

The institution acknowledges and endorses their intent to propose and participate in the NASA Human Exploration Rover Challenge 2026, should they be accepted.

Sincerely,

Ximena Peres Arenas
**Head of Institution**
**Universidad Católica Boliviana "San Pablo" - Sede La Paz**

**Bibliography**

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